

Project Proposal

SOS Emergency Response System

Component 1: Tourist SOS PWA + Police Dashboard

Field	Details
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1. Executive Summary

This proposal describes the design and implementation of an SOS Emergency Response System as part of the Tour Ceylon unified tourism platform. The system addresses a critical gap for international visitors in Sri Lanka: the lack of a fast, trusted, location-aware channel to reach police assistance during emergencies.

The solution comprises two web applications — a Tourist Progressive Web App (PWA) for one-tap emergency alerts and a Police Dashboard for station officers to acknowledge, dispatch, communicate with, and close incidents. Both applications share a unified Flask backend and PostgreSQL database with the Travel itinerary component.

2. Background & Motivation

- Sri Lanka welcomed over 1.5 million tourists annually pre-pandemic, with numbers recovering strongly. Visitors often lack knowledge of local emergency numbers, nearest police desks, or how to communicate location under stress.
- Existing emergency channels (119 hotline) do not automatically share GPS coordinates or route incidents to the geographically nearest dispatchable police station.
- Tour operators and hotels are not automatically notified when a registered tourist triggers an emergency, delaying family or staff awareness.
- The Tour Ceylon platform already serves tourists for trip planning; extending it with an integrated SOS module creates a single account experience from planning through safety.

3. Problem Statement

Foreign tourists in Sri Lanka face difficulty obtaining timely police assistance during medical emergencies, theft, assault, accidents, or harassment. They may not know which police station covers their location, cannot easily share accurate GPS coordinates by phone, and have no digital workflow to track whether help has been acknowledged or dispatched. Police stations lack a dedicated digital queue for tourist SOS incidents with location, tourist profile, and communication tools in one place.

4. Project Objectives

#	Objective	Success Indicator
O1	Enable one-tap SOS alerts with GPS from a mobile-friendly PWA	Tourist triggers incident in <3 taps with lat/lng captured
O2	Route each incident to the single nearest dispatchable police station	Haversine distance computed; station_id stored on incident
O3	Provide police workflow: Acknowledge → Dispatch → Close	Status timestamps recorded; tourist UI reflects updates
O4	Support incident chat between tourist and assigned station	Text, image, and voice messages stored per incident
O5	Integrate with Travel app login (no duplicate registration)	JWT bridge-travel endpoint exchanges Travel token for SOS session
O6	Export closed incident PDF reports for police records	ReportLab PDF downloadable after Close status
O7	Protect sensitive tourist PII with encryption at rest	Fernet encryption on phone, passport/NIC, emergency contact

5. Proposed Solution

The SOS system is split into three tiers: Tourist PWA, Police Dashboard, and Shared Backend.

5.1 Tourist PWA (port 5175)

- Six-type SOS bar: general, medical, assault, theft, accident, harassment
- One-shot browser geolocation at trigger — no continuous tracking (privacy by design)
- Real-time workflow tracker: Open → Acknowledged → Dispatched → Closed
- Incident chat with assigned police station
- Call nearest hospital via Haversine lookup + tel: dial link
- Installable PWA for mobile; Travel app handoff via JWT bridge

5.2 Police Dashboard (port 5176)

- Station-scoped officer login — each station sees only its own incidents
- Live incident queue with audio alert on new open cases
- Leaflet map showing tourist GPS pin and distance to station
- Workflow buttons: Acknowledge, Dispatch, Close
- Analytics charts: incident volume, response times, type breakdown
- PDF export for closed incidents

5.3 Shared Backend (port 5002)

- Unified Flask monolith shared with Travel component
- REST API with JWT authentication (tourist role vs officer role)
- Haversine routing service for nearest station and nearest hospital
- Notification service (console stub; Twilio/SendGrid optional)
- PostgreSQL persistence for tourists, incidents, stations, chat, hospitals

6. Methodology

Emergency routing uses deterministic Haversine great-circle distance — not machine learning. This approach is transparent, explainable, fast, and appropriate for nearest-station dispatch.

Phase	Activity	Deliverable
1. Requirements	Stakeholder analysis, user stories for tourist and officer flows	Requirements document + wireframes
2. Data collection	Import police station and hospital coordinates from Excel datasets	Seeded PostgreSQL tables
3. Backend API	Flask routes, JWT auth, routing, notify, PDF export	REST API on port 5002
4. Tourist PWA	React + Vite PWA with SOS trigger, chat, workflow	frontend/tourist on port 5175
5. Police Dashboard	React + Leaflet map, queue, stats, workflow	frontend/police on port 5176
6. Integration	Travel JWT bridge, shared database, end-to-end demo	Unified Tour Ceylon platform
7. Testing	Manual demo flow, routing validation, security review	Demo script + panel documentation

7. Technology Stack

Layer	Technology	Purpose
Backend	Flask 3.0.3	Unified REST API
Tourist UI	React 18 + Vite 5 PWA	Mobile-friendly SOS app
Police UI	React 18 + Vite + Leaflet	Station dashboard with map
Database	PostgreSQL (travel_app)	Incidents, tourists, stations, chat
Auth	Flask-JWT-Extended	Role-based tokens (tourist, officer)
Encryption	Fernet (cryptography)	PII at rest
PDF	ReportLab	Closed incident reports
Charts	Recharts	Police analytics dashboard
Routing	Haversine (Python math)	Nearest station/hospital

8. Data Requirements

Dataset	Source	Records	Usage
Police stations	Sri_Lanka_Police_Emergency_with_Coordinates.xlsx	Real desks	Routing targets
Hospitals	Sri_Lanka_Government_Hospitals_with_Coordinates.xlsx	Gov + private	Nearest hospital lookup
Officer credentials	station_credentials.csv (generated)	1 per station	Police login
Demo tourists	seed_sos_data.py	Simulated	Presentation/testing

9. Scope

In scope

- Tourist registration, login, and Travel JWT bridge
- Six-type SOS trigger with one-shot GPS
- Nearest-station routing and incident creation
- Police acknowledge/dispatch/close workflow
- Incident chat (text, image, voice)
- Nearest hospital dial lookup
- Police analytics and PDF export
- Fernet encryption of sensitive fields

Out of scope (future work)

- Continuous live GPS tracking after SOS trigger
- Automatic hospital dispatch or ambulance routing
- WebSocket real-time updates (polling used instead)
- Integration with national 119 emergency switchboard
- Multi-language tourist interface
- Automated unit/integration test suite

10. Expected Outcomes & Impact

- Tourists gain a trusted digital SOS channel integrated with their trip planning account.
- Police stations receive structured incident queues with GPS, tourist profile, and chat.
- Response workflow is auditable via timestamps and exportable PDF reports.
- Privacy is preserved through one-shot location capture and encrypted PII.
- The system demonstrates practical use of geospatial routing without ML complexity.
- Platform readiness for future SMS/email notification via Twilio/SendGrid configuration.

11. Proposed Timeline

Week	Milestone
1–2	Requirements, data import, database schema design
3–4	Backend API: auth, incidents, routing, hospitals
5–6	Tourist PWA: SOS trigger, workflow, chat, hospital dial
7–8	Police Dashboard: queue, map, workflow, stats, PDF

Week	Milestone
9	Travel JWT bridge integration
10	Testing, documentation, panel preparation

12. Conclusion

The SOS Emergency Response System provides a practical, privacy-conscious solution for tourist safety in Sri Lanka. By combining a one-tap mobile PWA, deterministic nearest-station routing, and a station-scoped police dashboard, the system closes a real gap in digital tourism infrastructure. As Component 1 of Tour Ceylon, it integrates seamlessly with the Travel itinerary planner, offering tourists a unified platform from trip planning through emergency assistance.

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